

Optimal power disaggregation among heterogeneous resources for providing frequency regulation

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- 02** | **The proposed framework**
- 03** | **Optimal Bidding Model with Embedded Power Allocation**
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- 05** | **Case Study**



Background

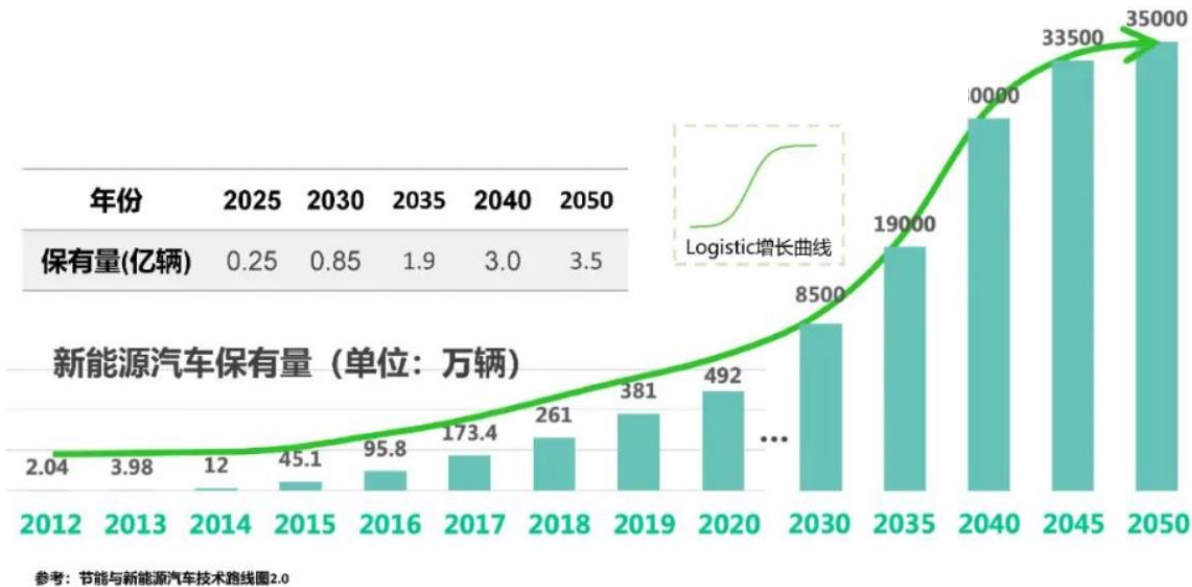
The novel power system needs more flexible resources!



August 2022: Power restriction in Sichuan. Chengdu residents using ice blocks to cool down after power restriction/Dimmed subway lights.

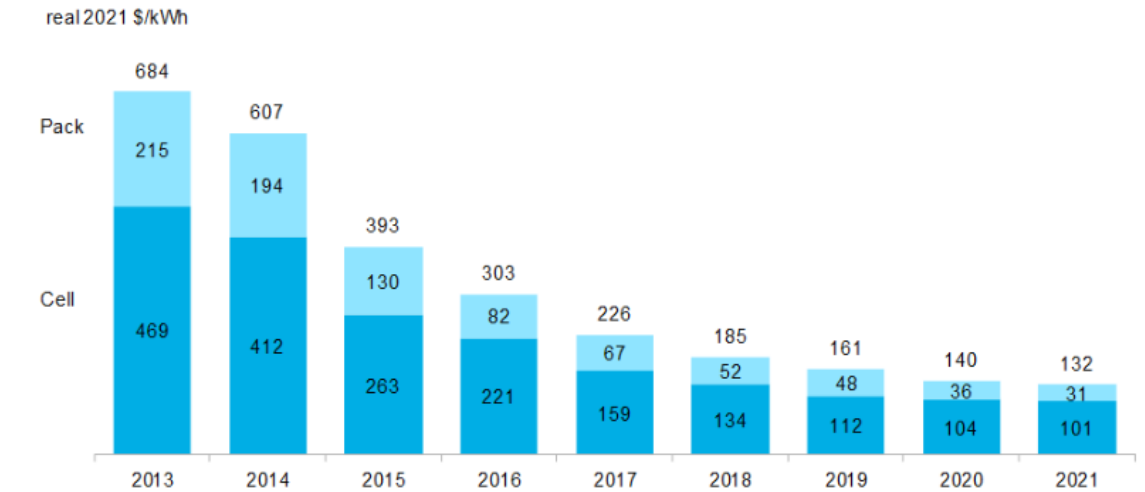
Background

V2G is becoming a reality!



In 2021, the number of electric vehicles in Beijing exceeded 500,000, able to support one-fourth of the maximum load of Beijing.

Figure 1: Volume-weighted average pack and cell price split



Source: BloombergNEF.

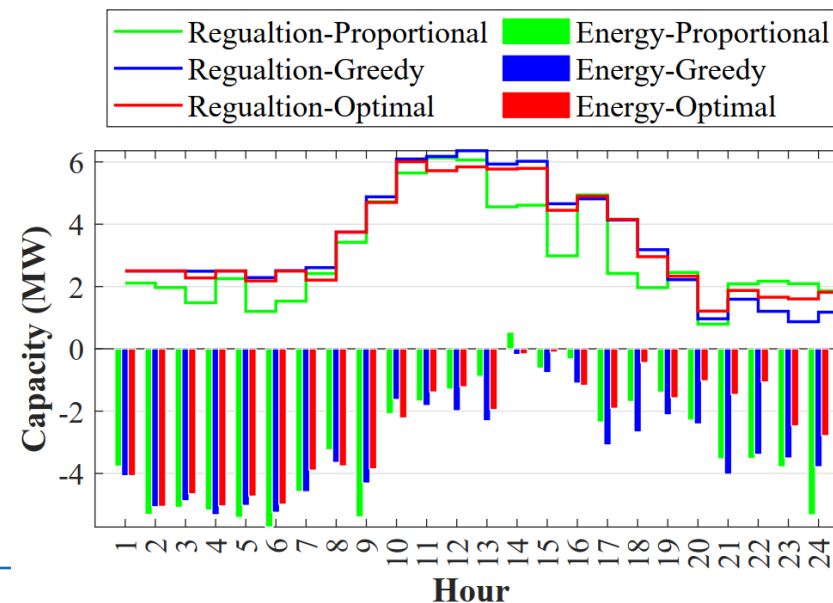
From 2014 to 2021, the average cost of power batteries has dropped by more than 75%.

Background

Due to the limited capacity of a single EV, EV aggregators (**EVA**s) usually need to aggregate a large number of EVs to meet the threshold for entering electricity market.

EVA-led charging management mode: three links

- the EVA **bidding** in the market: **deciding the regulation capacity**



bids of the VPP in the electricity market.

Background

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EVA-led charging management mode: three links

- the EVA **bidding** in the market: **deciding the regulation capacity**
- the EVA **allocating** (dis)charging power to EVs

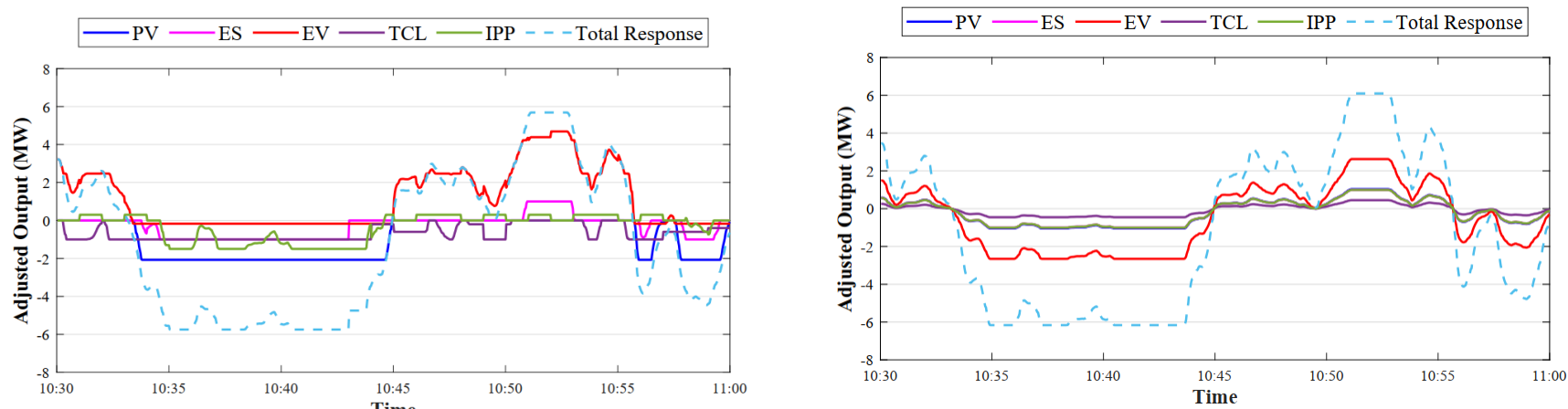


Figure 4: Power aggregation results for deploying regulation at 10 a.m., July 21: (a) Optimal disaggregation. (b) Proportional disaggregation.

Background

Due to the limited capacity of a single EV, EV aggregators (**EVA**s) usually need to aggregate a large number of EVs to meet the threshold for entering electricity market.

EVA-led charging management mode: three links

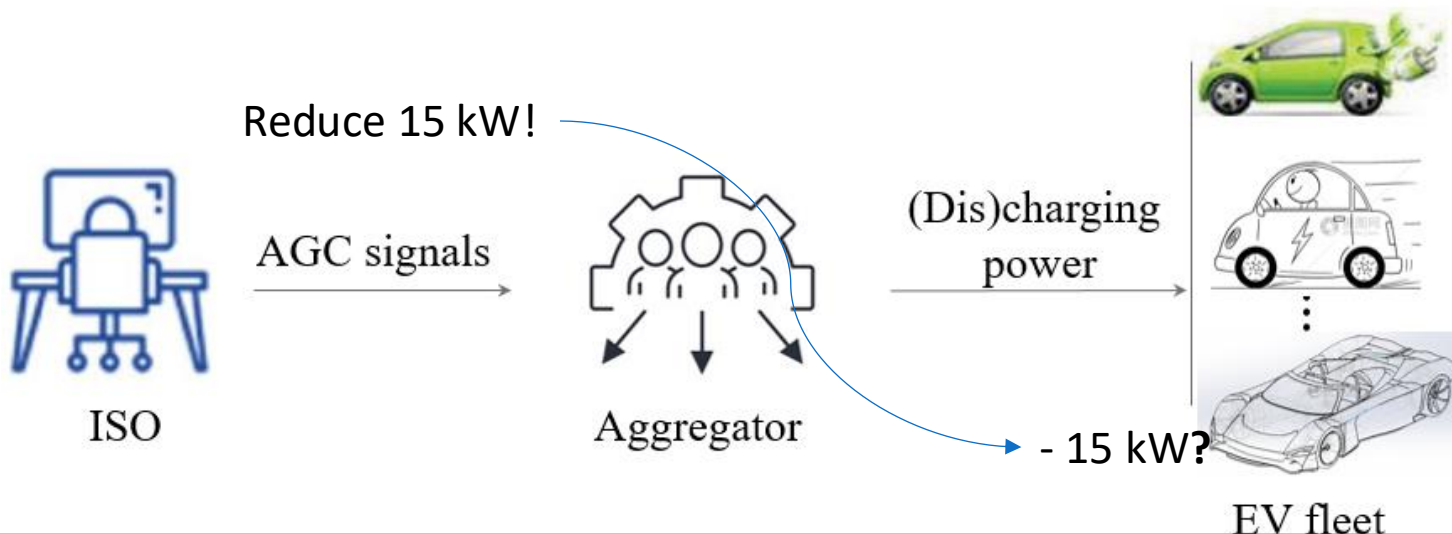
- the EVA **bidding** in the market: **deciding the regulation capacity**
- the EVA **allocating** (dis)charging power to EVs
- the **revenue distribution** between the EVA and EV owners

The three links are **coupled but often regarded as decoupled** and optimized separately for complexity issues. A **practical framework for co-optimizing EVA bidding and power allocation** is challenging!

Motivation

Why co-optimizing?

a. **In the EVA bidding problem**, it is necessary to consider the **power allocation strategy** in regulation deployment because different allocation strategies will induce different costs, especially when the EVs feature **heterogeneous degradation costs, charging efficiencies, and other characteristics**.



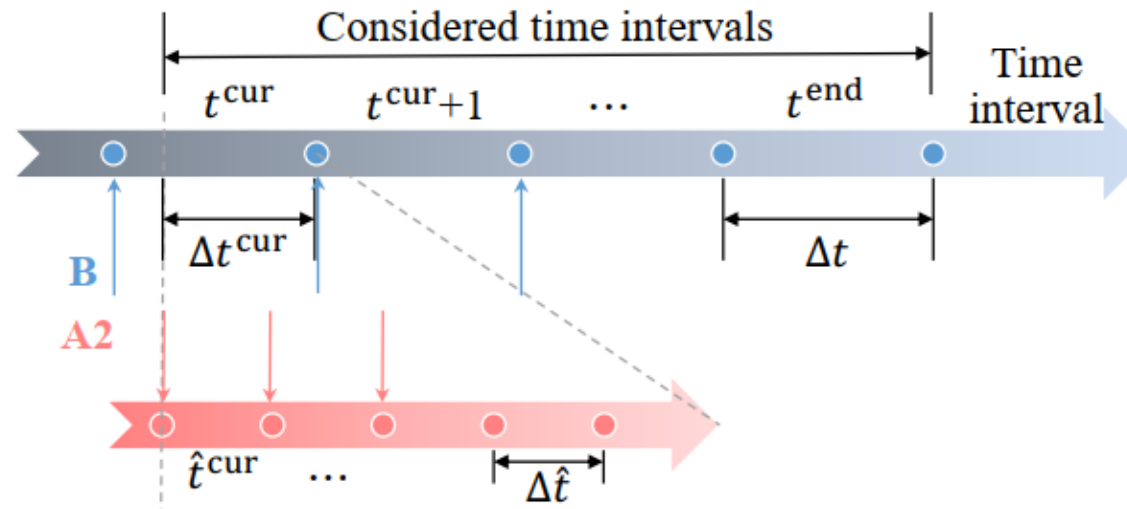
Intuition: allocate discharging power to the EVs with lower degradation costs!

What if always allocating discharging power to the EVs with lower degradation costs?

Motivation

Why co-optimizing?

b. As a result of the market operation process, **bidding and power allocation are alternating and process coupled**. The bids for a certain time interval determine the regulation power to be allocated in the interval, and the power allocation results determine the feasible bidding range for the next time intervals by affecting the SOC of the EVs.

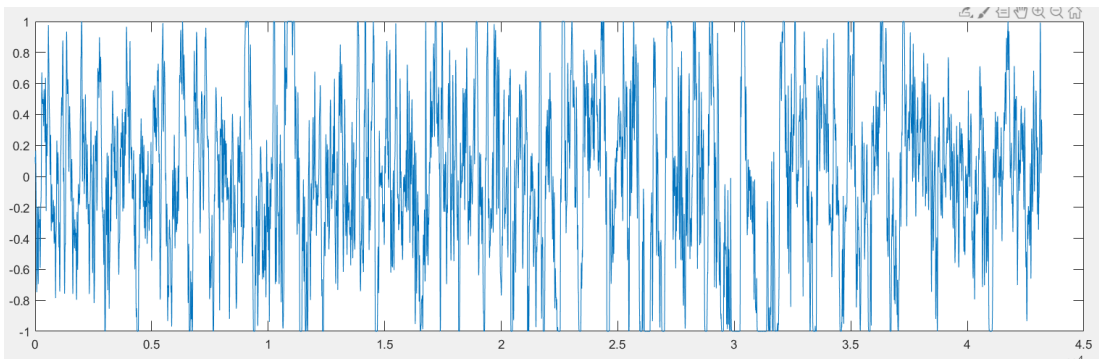


Motivation

Why challenging?

Calculation time: Following the regulation signals that are issued every few seconds means that the (dis)charging power of thousands of EVs needs to be determined online.

Uncertainty: Regulation (AGC) signals are not known in advance.



RegD signal form PJM, 18.07.2022

How to online solve stochastic optimization with a large number of scenarios?



Literature Review

TABLE I
COMPARISON OF RELEVANT LITERATURE

Ref. on EVA Bidding	Consideration for Power Allocation	Ref. on Power Allocation	Consideration for Bidding Profit
[11]–[15], [17], [18]	Allocated Proportional to the allocated regulation capacity	[9], [19], [20]	Not considered or not related
[16]	Evenly allocated to the EVs	[10]	Maximizing the regulation capacity at the next time interval
Our bidding model	Optimized simultaneously	Our power allocation	Optimal for the bidding profit

$$FP_i = (RRS/R_R) \cdot \frac{RsRP_i + PD_i}{RsRP_i + PD_i}$$

Proportionally allocation.

$$w_i^{Er} = 1 - \left(\frac{E_i^r}{\max_i (E_i^{bat})} \right)^{\frac{1}{y}} \quad \forall i$$

$$w_i^{Tr} = \left(\frac{T_i^r}{\max_i (T_i^{dep} - T_i^{arr})} \right)^{\frac{1}{z}} \quad \forall i$$

Heuristic weights.

$$w_i = (w_i^{Er})^{x_1} \cdot (w_i^{Tr})^{x_2}$$

There is no research that reasonably models the **coupling** and provides a practical framework that meets the **time** constraints required for regulation deployment. This problem exists in aggregating EVs, energy storages, HVAC and other resources.



Contribution

1. We propose a framework to **co-optimize the bidding and power allocation** of an EVA in the regulation market. The bidding model is formulated as stochastic optimization with embedded power allocation in discretized regulation signal scenarios.
2. To meet the computation time requirement for regulation deployment, we propose a power allocation model using the Lagrange multipliers from the bidding model, which **can be calculated online**. This determines the allocated power that corresponds to the optimal solution of the bidding model.
3. The case study verifies the effect of considering the coupling between bidding and power allocation on **improving the EVA profit and reducing EV battery degradation costs**.

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The proposed framework

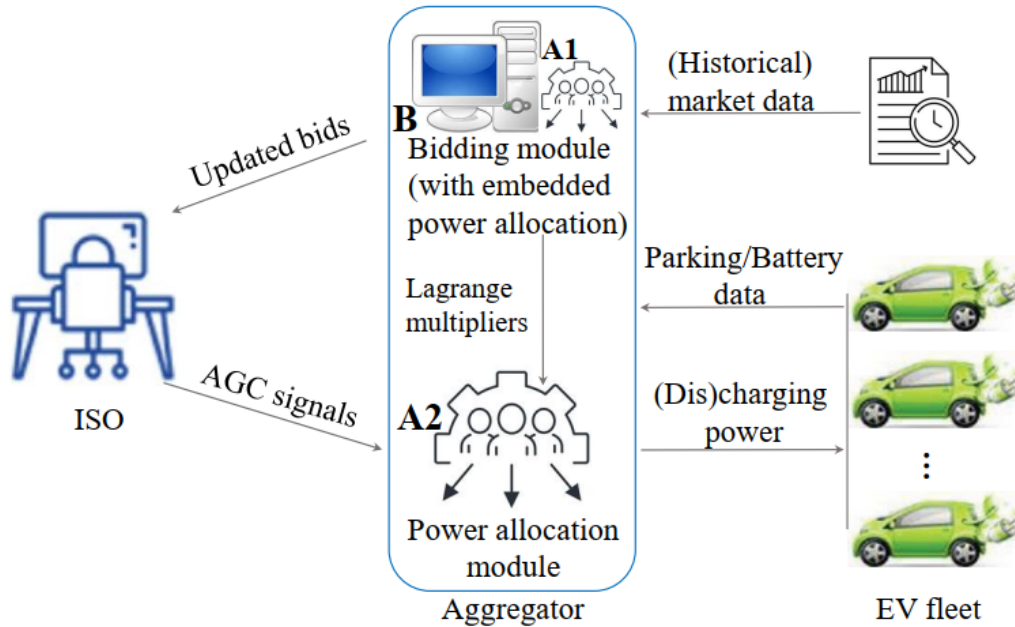


Fig. 1. Overview of the proposed framework for EVA bidding and power allocation co-optimization.

1. Real-time frequency regulation market.
2. EVA Contracts with EV owners.
3. Bidding module (B)
4. Power allocation module (A2)

The proposed framework

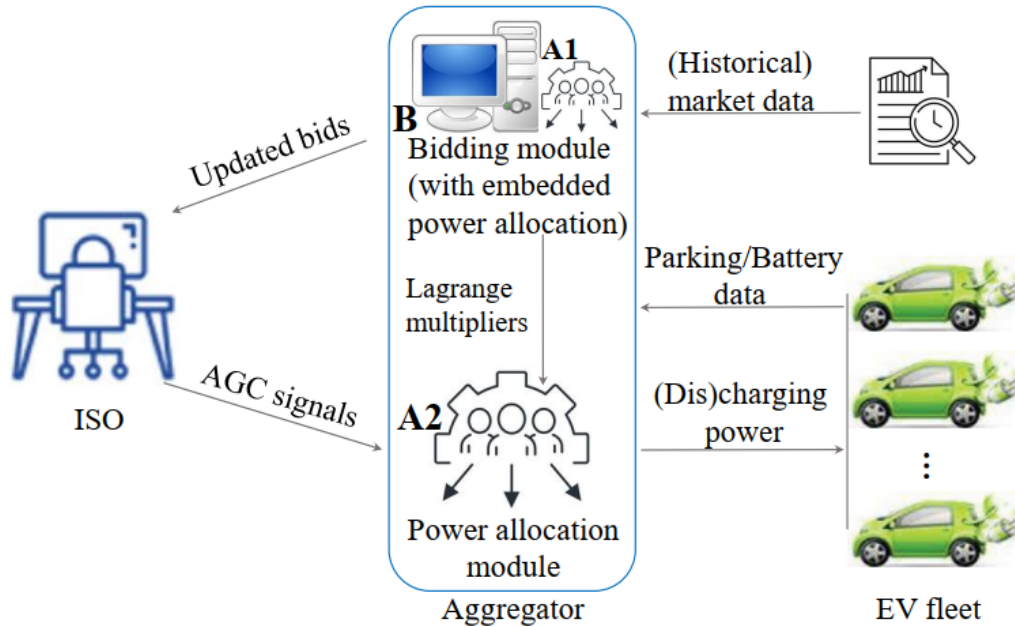


Fig. 1. Overview of the proposed framework for EVA bidding and power allocation co-optimization.

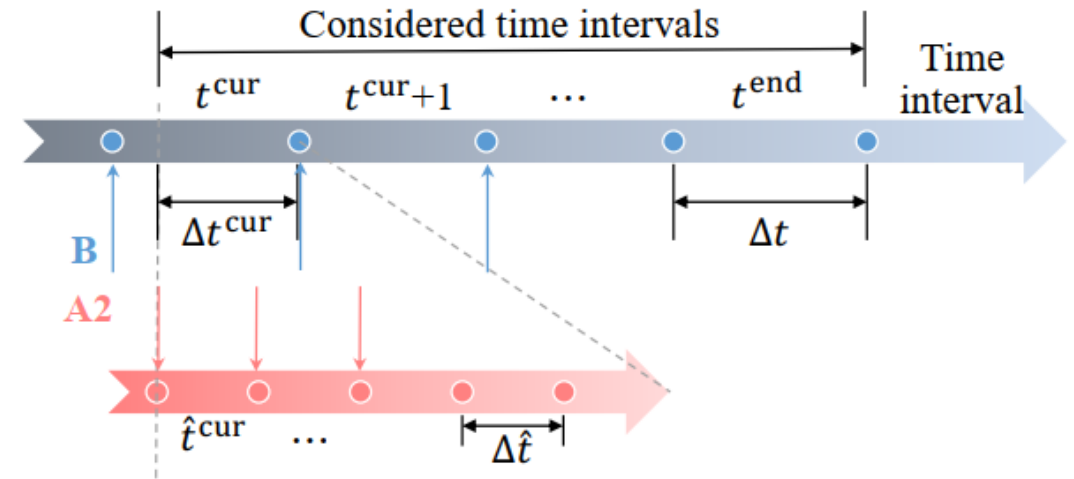


Fig. 2. Timeline of the proposed framework: (B) Update the bids (periodically, e.g., every 30 minutes); (A2) Allocate the power for deploying regulation to EVs (upon receiving an AGC signal, e.g., every 2 seconds).

1. Real-time frequency regulation market.
2. EVA Contracts with EV owners.
3. Bidding module (B)
4. Power allocation module (A2)

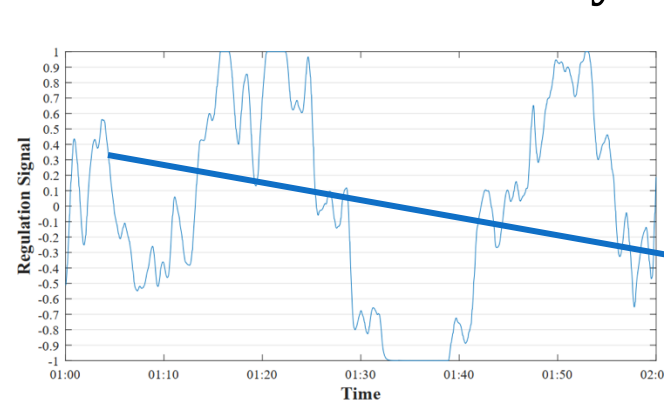
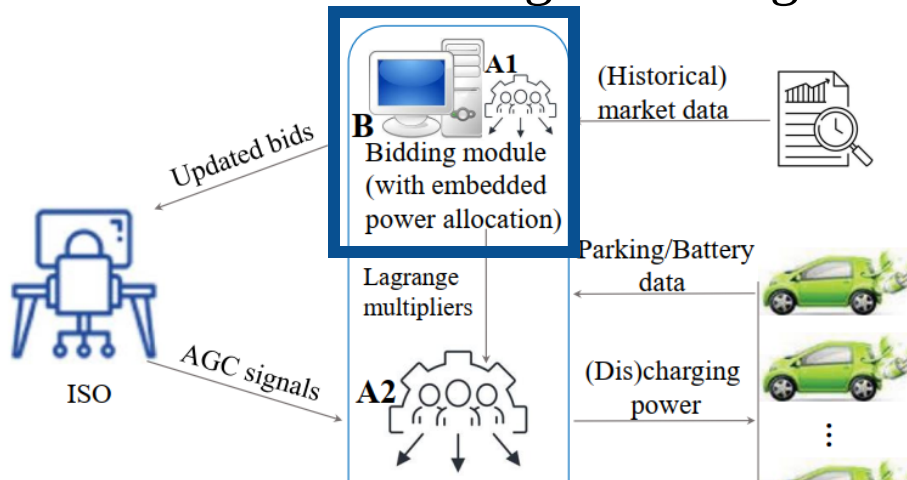
The proposed framework

Bidding module (B)

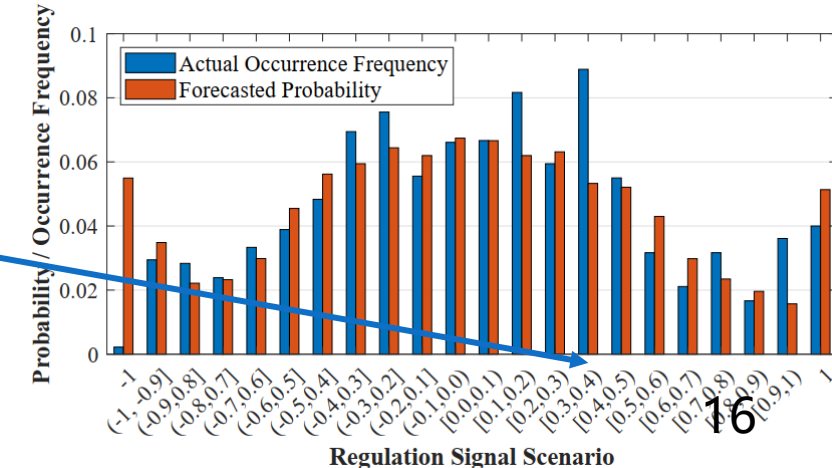
Data preparation: the EVA collects the arriving and leaving time, current SOC, charging needs, and battery parameters of the EVs and generates forecasts about prices and regulation signals according to historical market data.

Optimal bidding: based on these parameters, the bidding module outputs the latest optimal bids, and the Lagrange multipliers can be obtained without additional computational overhead.

Power allocation submodule (A1) embedded in the bidding module (B): the power allocation in each regulation signal scenarios is **simultaneously optimized**.



actual regulation signals



The proposed framework

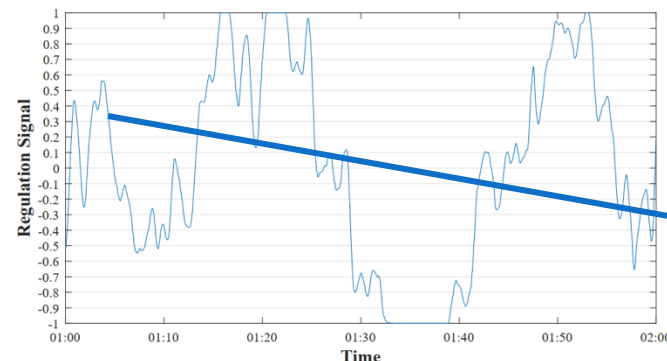
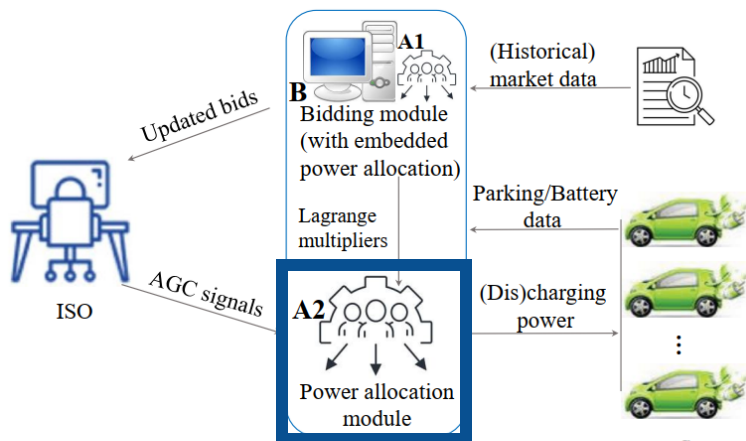
Optimal power allocation module (A2)

We design a power allocation module (A2) to quickly decide the (dis)charging power of the EVs in regulation deployment.

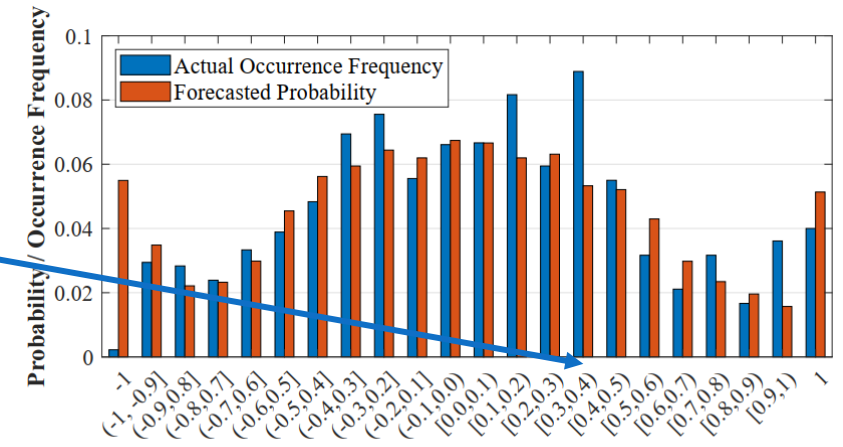
Why A2?

The power allocation results (A1) in the bidding model only correspond to the discretized regulation signal scenarios (e.g., $[0, 0.1)$, ..., $[0.9, 1)$,...), while the actual regulation signal may take any value in its nearly continuous range (e.g., 0.9998).

Therefore, the bidding model cannot be directly used to respond to regulation signals.



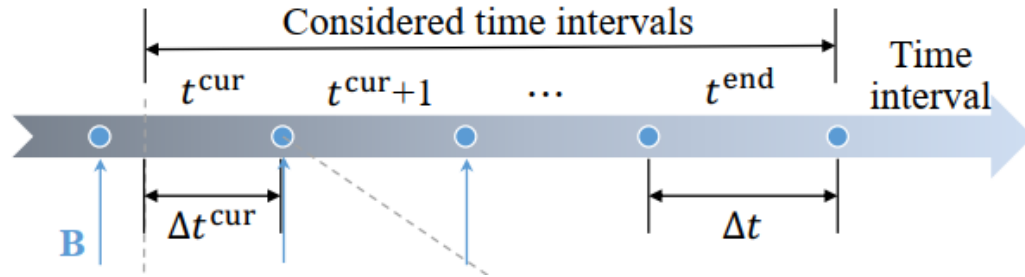
actual regulation signals



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Bidding Model



A. Profit Model

The EVA aims to maximize $Profit(t^{cur})$, the sum of its profits for future time intervals from the first (current) time interval t^{cur} to the end t^{end} , as formulated in (1).

$$\begin{aligned}
 Profit(t^{cur}) = & \sum_{s \in S} \pi_s \sum_{t \in T} (Income_{s,t}^e + Income_{s,t}^{cap} \\
 & + Income_{s,t}^{mil} + Income_{s,t}^{ch} \\
 & + \sum_{\hat{s} \in \hat{S}} \pi_{t,\hat{s}} (Income_{s,t,\hat{s}}^{deploy} - Cost_{t,\hat{s}}^{deg}))
 \end{aligned} \quad (1)$$

s/S

Index/set of price scenarios.

$\hat{s}/\hat{S}$

Index/set of regulation signal scenarios.

$\pi_{t,\hat{s}}$

Probability of regulation signal scenario $\hat{s}$ at time interval t .

π_s

Probability of price scenario s .

$$Cost_{t,\hat{s}}^{deg} = \sum_{i \in I} \sum_{k \in K} Pr_{i,k}^{deg} \cdot P_{t,\hat{s},i,k}^{dis} \cdot \Delta t$$

$Pr_{i,k}^{deg}$

Degradation cost rate at power segment k of EV i (\$/kWh).

$P_{t,\hat{s},i,k}^{dis(ch)}$

Discharging (charging) power at segment k of EV i at time interval t in regulation scenario $\hat{s}$ (kW).

- We use the degradation model for EV batteries* where the battery degradation is based on the **depth of discharge (DOD)** and can be **linearly approximated by discharge segments**.

Bidding Model

B. Constraints

In the bidding model, the EVA needs to meet the technical and charging need constraints of the EVs (7)-(14).

$$0 \leq R_t, \forall t \in T \quad (7)$$

$$P_t + R_t \cdot \delta_{\hat{s}} - \sum_{i \in I} \sum_{k \in K} (P_{t,\hat{s},i,k}^{\text{dis}} - P_{t,\hat{s},i,k}^{\text{ch}}) = 0 : \lambda_{t,\hat{s}}^{\text{bal}}, \quad \forall t \in T, \hat{s} \in \hat{S} \quad (8)$$

$$\sum_{i \in I} (E_{t,i} - E_{t,i}^{\min}) \cdot \eta_i^{\text{dis}} \geq R_t \cdot \Delta t^{\text{req}} + P_t \cdot \Delta t, \quad \forall t \in T \quad (9)$$

$$\sum_{i \in I} (-E_{t,i} + E_{t,i}^{\max}) / \eta_i^{\text{ch}} \geq R_t \cdot \Delta t^{\text{req}} - P_t \cdot \Delta t, \quad \forall t \in T \quad (10)$$

$$0 \leq P_{t,\hat{s},i,k}^{\text{dis(ch)}} \leq u_{t,i} \cdot \overline{P_{i,k}^{\text{dis(ch)}}} : \underline{\mu_{t,\hat{s},i,k}^{\text{dis(ch)}}}, \overline{\mu_{t,\hat{s},i,k}^{\text{dis(ch)}}}, \quad \forall t \in T, \hat{s} \in \hat{S}, i \in I, k \in K \quad (11)$$

$$E_i^{\text{cur}} - E_{t_{\text{cur}},i} = 0, \forall i \in I \quad (12)$$

$$E_{t,i} + \sum_{\hat{s} \in \hat{S}} \pi_{t,\hat{s}} \sum_{k \in K} (P_{t,\hat{s},i,k}^{\text{ch}} \cdot \eta_i^{\text{ch}} - P_{t,\hat{s},i,k}^{\text{dis}} / \eta_i^{\text{dis}}) \cdot \Delta t - E_{t+1,i} = 0 : \lambda_{t,i}^{\text{E}}, \quad \forall t \in T, i \in I \quad (13)$$

$$E_{t,i}^{\min} \leq E_{t,i} \leq E_{t,i}^{\max}, \quad \forall t \in T, i \in I \quad (14)$$

P_t

Energy bid at time interval t (MW).

R_t

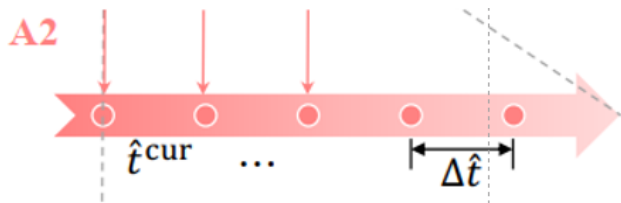
Regulation capacity bid at time interval t (MW).

- Net power of the EVs follows the regulation signal in each regulation signal scenario.
- The expected change of EV i in SOC across the time intervals.
- The entire model, formulated as a **stochastic linear programming problem**, can be efficiently solved by commercial solvers.

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Power Allocation Model



A. Power Allocation Model

At time $\hat{t}^{\text{cur}} \in t^{\text{cur}}$, upon receiving a regulation signal $\delta_{\hat{s}}$ from the ISO, the EVA runs the following power allocation problem (17)-(23):

$$\min. \text{Cost}_{\hat{t}^{\text{cur}}, \hat{s}} + M \cdot (\Delta P_{\hat{t}^{\text{cur}}, \hat{s}}^{\text{dis}} + \Delta P_{\hat{t}^{\text{cur}}, \hat{s}}^{\text{ch}}) \quad (17)$$

$$\text{s.t. } \text{Cost}_{\hat{t}^{\text{cur}}, \hat{s}} = \sum_{i \in I} \sum_{k \in K} (P_{i,k}^{\text{deg}} \cdot P_{\hat{t}^{\text{cur}}, \hat{s}, i, k}^{\text{dis}} + \lambda_{t^{\text{cur}}, i}^{\text{E}} \cdot (P_{\hat{t}^{\text{cur}}, \hat{s}, i, k}^{\text{ch}} \cdot \eta_i^{\text{ch}} - P_{\hat{t}^{\text{cur}}, \hat{s}, i, k}^{\text{dis}} / \eta_i^{\text{dis}})) \cdot \Delta \hat{t} \quad (18)$$

$$P_{t^{\text{cur}}} + R_{t^{\text{cur}}} \cdot \delta_{\hat{s}} - \sum_{i \in I} \sum_{k \in K} (P_{\hat{t}^{\text{cur}}, \hat{s}, i, k}^{\text{dis}} - P_{\hat{t}^{\text{cur}}, \hat{s}, i, k}^{\text{ch}}) - (\Delta P_{\hat{t}^{\text{cur}}, \hat{s}}^{\text{dis}} - \Delta P_{\hat{t}^{\text{cur}}, \hat{s}}^{\text{ch}}) = 0 : \lambda_{\hat{t}^{\text{cur}}, \hat{s}}^{\text{bal}} \quad (19)$$

⋮

- The second term in the objective function is a penalty term, which forces the EVA to follow the regulation signal as closely as possible.
- The cost is calculated from the battery **degradation cost that is immediately generated** by EV discharge in addition to **the impact on future EVA profit accounted for through the Lagrange multiplier**, which will be analyzed later.

Power Allocation Model

B. Optimality of Power Allocation

In this paper, the optimality of the power allocation problem refers to the fact that its output (allocated power) **corresponds to the optimal power allocation** given by the bidding problem.

Intuitively, the Lagrange multiplier of the SOC constraint $\lambda_{t,i}^E$ represents the **shadow price of SOC** for the profit of EVA bidding, i.e., the cost of increasing a unit SOC of EV i at t .

By considering the **impact on future EVA profit through $\lambda_{t,i}^E$ together with the immediately induced** degradation cost, the solution to the power allocation problem maximizes EVA profit.

Bidding problem

B. Constraints

In the bidding model, the EVA needs to meet the technical and charging need constraints of the EVs (7)-(14).

$$0 \leq R_t, \forall t \in T \quad (7)$$

$$P_t + R_t \cdot \delta_s - \sum_{i \in I} \sum_{k \in K} (P_{t,\hat{s},i,k}^{\text{dis}} - P_{t,\hat{s},i,k}^{\text{ch}}) = 0 : \lambda_{t,\hat{s}}^{\text{bal}}, \quad \forall t \in T, \hat{s} \in \hat{S} \quad (8)$$

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$$E_i^{\text{cur}} - E_{t,i} = 0, \forall i \in I \quad (12)$$

$$E_{t,i} + \sum_{\hat{s} \in \hat{S}} \pi_{t,\hat{s}} \sum_{k \in K} (P_{t,\hat{s},i,k}^{\text{ch}} \cdot \eta_i^{\text{ch}} - P_{t,\hat{s},i,k}^{\text{dis}} / \eta_i^{\text{dis}}) \cdot \Delta t - E_{t+1,i} = 0 : \lambda_{t,i}^E, \quad \forall t \in T, i \in I \quad (13)$$

$$E_{t,i}^{\min} \leq E_{t,i} \leq E_{t,i}^{\max}, \quad \forall t \in T, i \in I \quad (14)$$

Power allocation problem

$$\min. \text{Cost}_{t,\hat{s}}^{\text{cur}} + M \cdot (\Delta P_{t,\hat{s}}^{\text{dis}} + \Delta P_{t,\hat{s}}^{\text{ch}}) \quad (17)$$

$$\text{s.t. } \text{Cost}_{t,\hat{s}}^{\text{cur}} = \sum_{i \in I} \sum_{k \in K} (P_{i,k}^{\text{deg}} \cdot P_{t,\hat{s},i,k}^{\text{dis}} + \lambda_{t,i}^E \cdot (P_{t,\hat{s},i,k}^{\text{ch}} \cdot \eta_i^{\text{ch}} - P_{t,\hat{s},i,k}^{\text{dis}} / \eta_i^{\text{dis}})) \cdot \Delta t \quad (18)$$

$$P_{t,\hat{s}}^{\text{cur}} + R_{t,\hat{s}} \cdot \delta_s - \sum_{i \in I} \sum_{k \in K} (P_{t,\hat{s},i,k}^{\text{dis}} - P_{t,\hat{s},i,k}^{\text{ch}}) = 0 : \lambda_{t,\hat{s}}^{\text{bal}}, \quad -(\Delta P_{t,\hat{s}}^{\text{dis}} - \Delta P_{t,\hat{s}}^{\text{ch}}) = 0 : \lambda_{t,\hat{s}}^{\text{bal}} \quad (19)$$

Power Allocation Model

B. Optimality of Power Allocation

In the mathematical meaning, introducing Lagrange multipliers of the bidding problem into the power allocation problem makes **the optimality conditions of the two problems correspond**. Our analysis is based on the following assumptions:

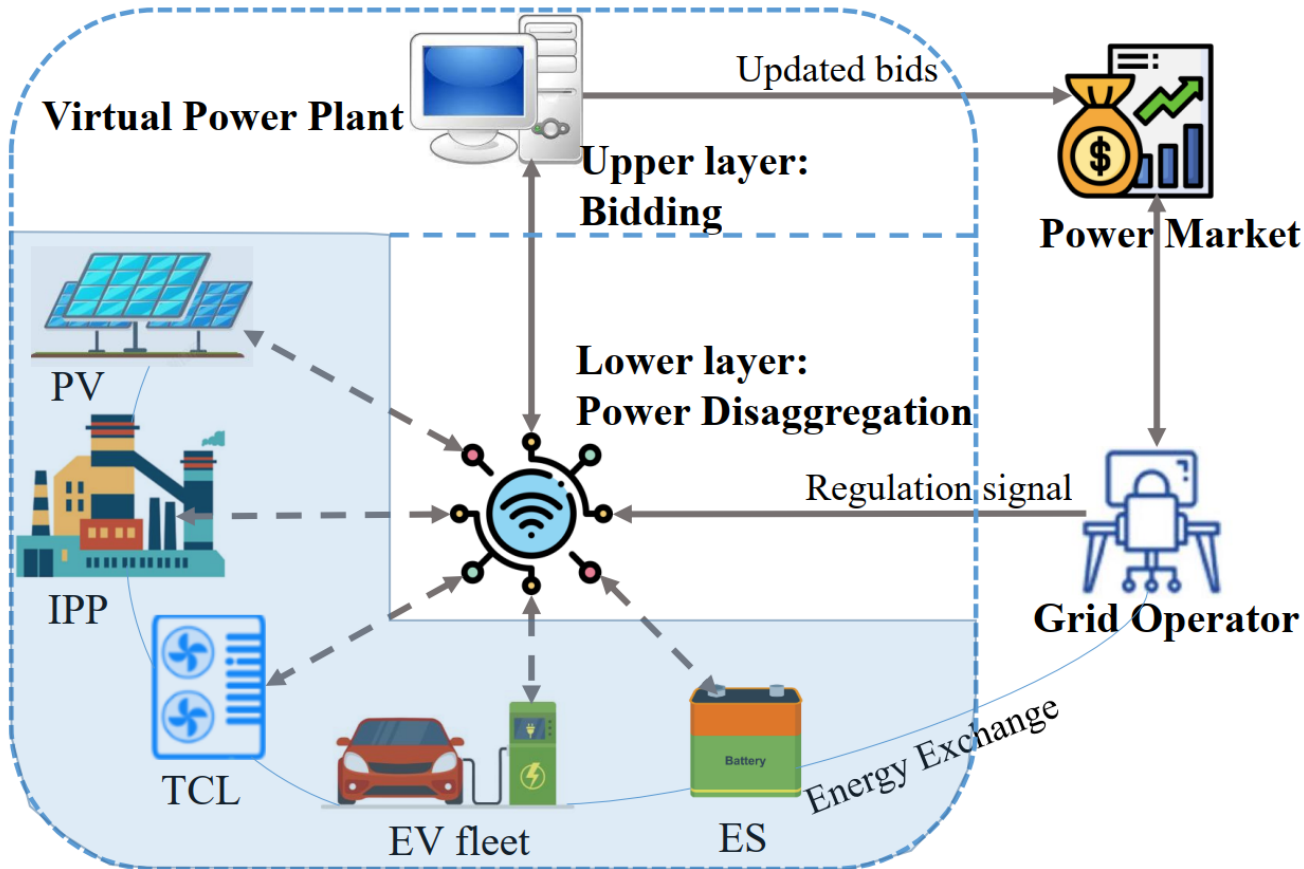
Assumption 1. *The bidding problem (16) and the power allocation problem (17) are strictly feasible.*

Assumption 2. *The actual regulation signals can be expressed by the regulation signal scenarios, i.e., $\forall \delta_{\hat{s}} \in [-1, 1]$, and we have $\hat{s} \in \hat{S}$ and $\pi_{t,\hat{s}} > 0, \forall t$.*

Assumption 3. *The occurrence frequency of the regulation signals is consistent with the forecasted probability.*

For Extension

We extended the optimal operation framework to VPPs composed of various distributed resources: establish a standardized model that can express the operating characteristics of multiple distributed resources and everything can be easily applied!



$p_{t,s,i} = p_{t,s,i}^{\text{dis}} - p_{t,s,i}^{\text{ch}}, \forall t, \forall s, \forall i$ $\underline{p}_{t,i}^{\text{dis(ch)}} \leq p_{t,s,i}^{\text{dis(ch)}} \leq \bar{p}_{t,i}^{\text{dis(ch)}}, \forall t, \forall s, \forall i$	Power limit
$\underline{e}_{t,i} \leq e_{t,i} \leq \bar{e}_{t,i}, \forall t, \forall i$	Energy limit
$e_{t+1,i} = \theta_i e_{t,i} + \sum_s \pi_{t,s} \sum_j \left(\eta_{ij}^{\text{ch}} p_{t,s,j}^{\text{ch}} - \frac{1}{\eta_{ij}^{\text{dis}}} p_{t,s,j}^{\text{dis}} \right) \Delta t$ $+ w_{t,i} \Delta t : \lambda_{t,i}^e, \forall t, \forall i$	Change of energy
$e_i^{\text{init}} - e_{t,i} = 0, t = 1, \forall i$	Initial energy
$\text{Cost}_t = \sum_i \sum_{s \in [S]} \pi_{t,s} \left(\text{Pr}_i^{\text{dis}} p_{t,s,i}^{\text{dis}} + \text{Pr}_i^{\text{ch}} p_{t,s,i}^{\text{ch}} \right) \Delta t, \forall t$	Response cost

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Case Study

A. Basic Data and Assumptions

- We used historical market data in July 2022 from PJM. We assumed that the prices can be accurately forecasted while the regulation-signal-related parameters in the bidding problem are forecasted results.

TABLE II
PARAMETERS OF THE THREE TYPES OF EVs

Type	Discharging Power Limit (kW)	Charging Power Limit (kW)	Initial SOC (kWh)	Required SOC (kWh)	Lower SOC Limit (kWh)	Upper SOC Limit (kWh)	Discharging Efficiency	Charging Efficiency	Degradation Factor (\$/kWh)
a	7.68	7.68	30	81	15	90	0.95	0.95	0.10
b	7.68	7.68	20	54	10	60	0.92	0.92	0.15
c	7.68	7.68	12	32	6	36	0.90	0.90	0.20

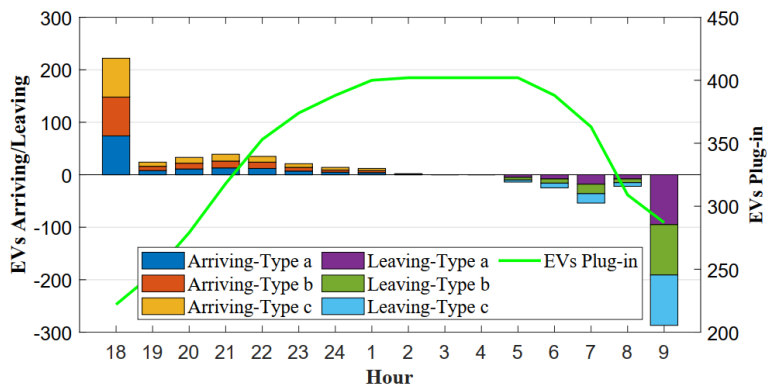


Fig. 3. Distribution of arriving and leaving time by the type of EVs.

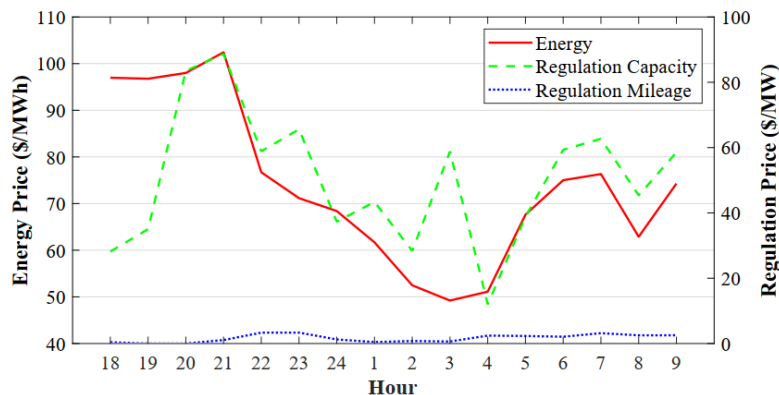


Fig. 4. Real-time market prices (17.07.2022 18:00 to 18.07.2022 9:00).

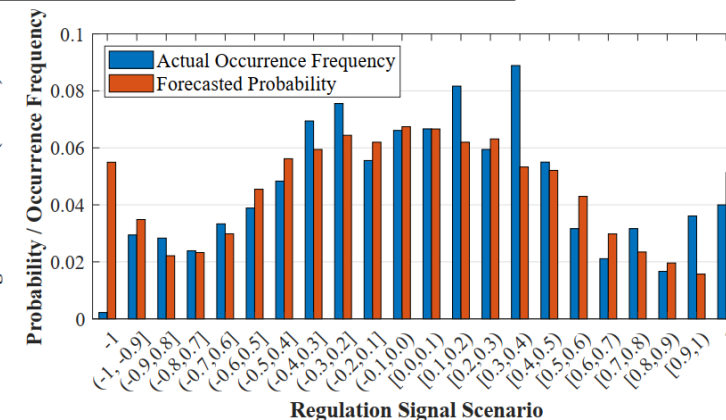


Fig. 5. Distribution of regulation signal scenarios at 6:00-7:00.

Case Study

B. Results and Comparison

The results using the following four bidding-power allocation methods are compared:

The proposed method: Power allocation is embedded in the bidding model (16), and the Lagrange-multiplier-based power allocation model (17) decides the power of the EVs in regulation deployment.

Proportional Allocation: In the bidding model, the EVA regulation capacity is allocated to the EVs, and the adjusted power of the EVs in regulation deployment is set proportional to the allocated capacity [15].

Heuristic Weight: Bidding as in [15]; degradation cost and charging priority are considered in power allocation. This is equivalent to replacing the Lagrange multipliers in (18) with the heuristic weights proposed in [20]. The recommended parameters are used, i.e., $(x_1, x_2, y, z) = (1, 1, 4, 5)$.

Minimum Degradation: Bidding as in [15]; power is preferentially allocated to EVs with lower degradation factors. This is equivalent to setting the Lagrange multipliers in (18) to be 0.

TABLE III
COMPARISON OF EVA PROFITS USING DIFFERENT METHODS

Bidding-Allocation Method	Income from the Market (\$)	Degradation Cost (\$)	EVA Profit (\$)
Proposed Method	1647 (-1.9%)	316 (-41.1%)	1331 (+16.7%)
Proportional Allocation	1679 0%	538 0%	1141 0%
Heuristic Weight	1672 (-0.4%)	446 (-16.9%)	1225 (+7.4%)
Minimum Degradation	1243 (-26.0%)	447 (-17.0%)	796 (-30.2%)

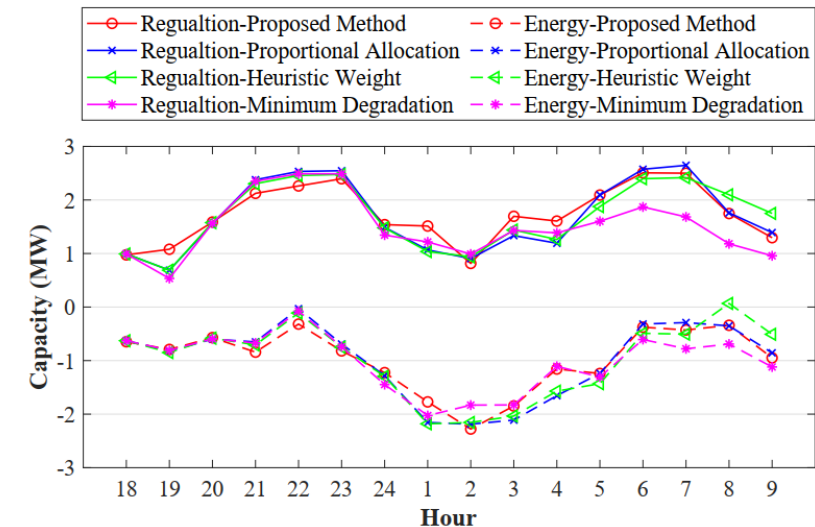


Fig. 4. Final bids of the EVA in the electricity market.

Case Study

EV211: Type a (low degradation cost), arrived at 18:00 on the 17th and left at 10:00 on the 18th).

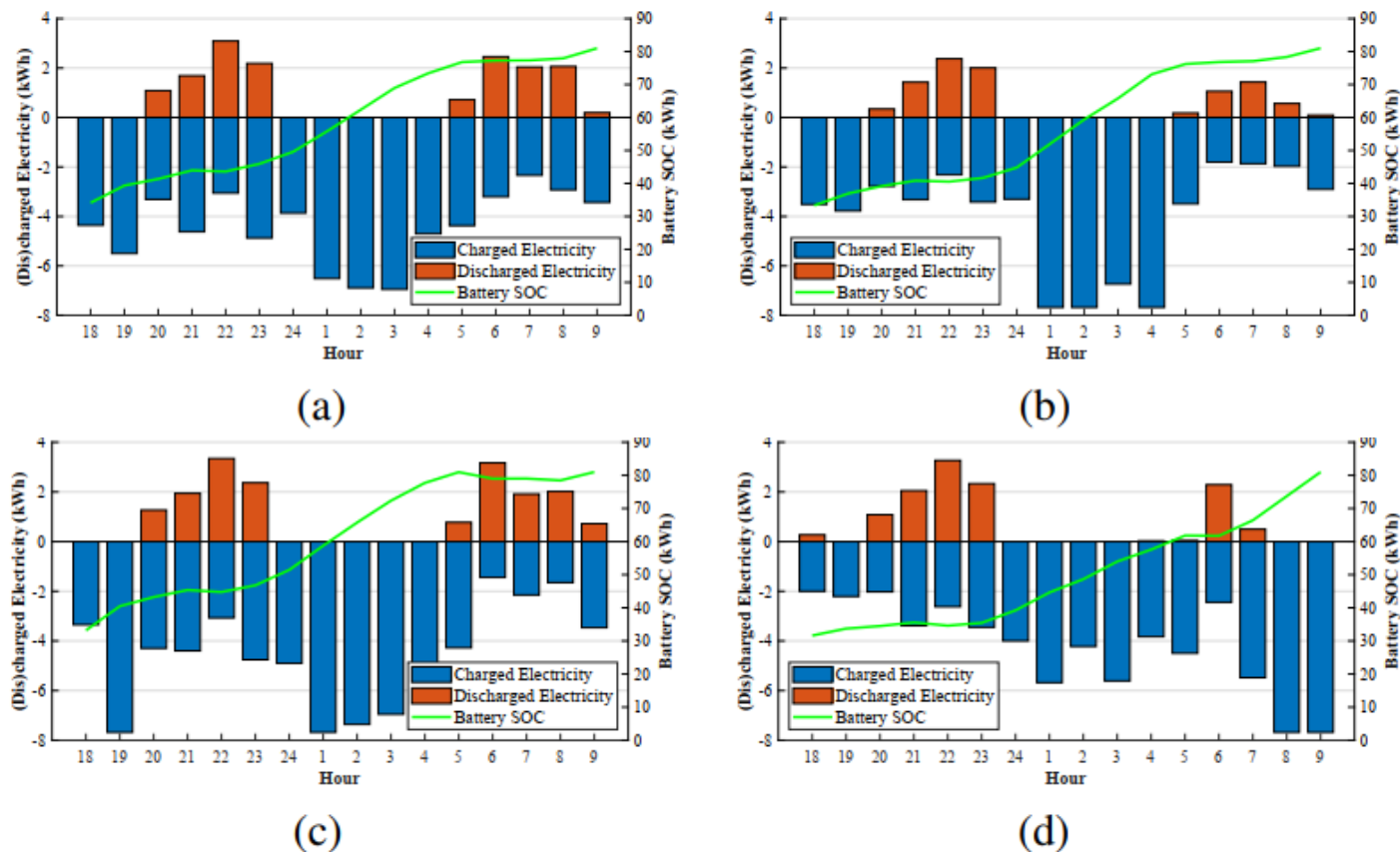


Fig. 7. Exchanged electricity with the grid and energy Curve of EV211: (a) The proposed method. (b) Proportional Allocation. (c) Heuristic Weight. (d) Minimum Degradation.

- Proportional Allocation: failing to utilize the low degradation cost of EV211.
- Minimum Degradation: ignoring the impact of discharging EV211 on future EVA profit.
- Heuristic Weight: partially reconciles the aim of reducing degradation costs and increasing future income.
- Proposed Method: optimal.

Case Study

C. Sensitivity Analysis

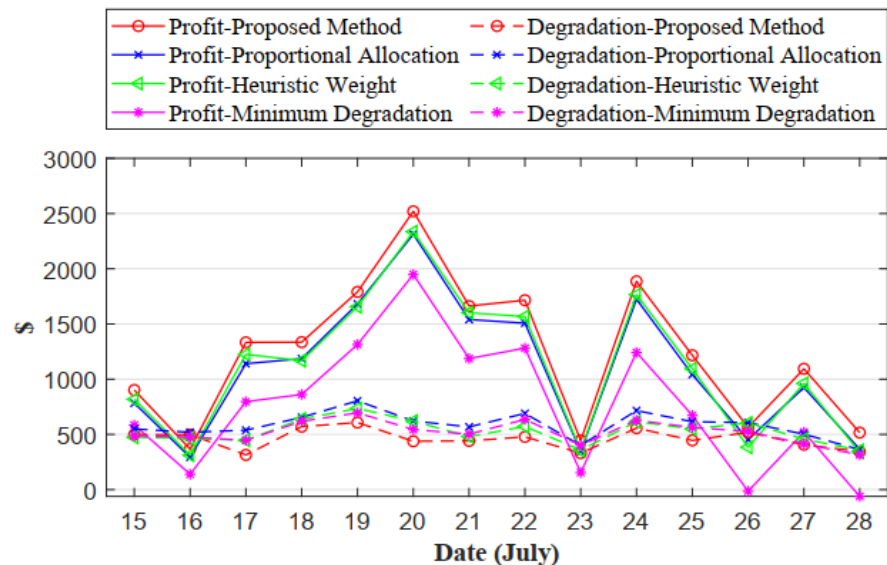


Fig. 8. EVA profits and battery degradation costs of July 15-28.

Compared to the Proportional Allocation method (blue), the increase ratio of profits using the proposed method (red) ranges between 6.5% and 41.3%, and the total profit of the 14 days is increased by 13.4%.

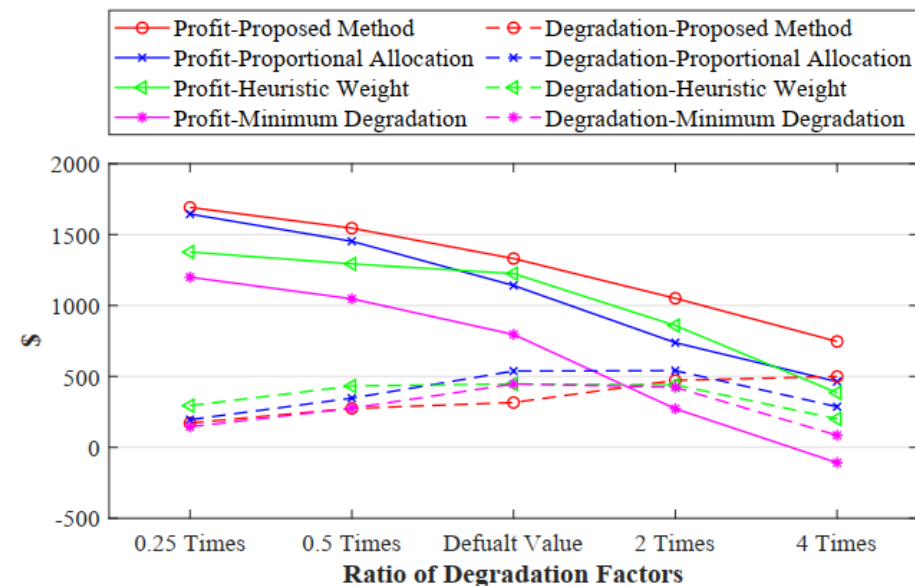


Fig. 9. EVA profits and battery degradation costs with different degradation factors.

Compared with the Proportional Allocation method (blue), when the degradation factors are 1, 2, and 4 times the default value, the EVA profits using the proposed method (red) are 16.7%, 42.4%, and 61.0% higher, respectively.

Case Study

C. Sensitivity Analysis

TABLE IV
IMPACT OF DIFFERENT UPDATE PERIODS

Update Period (min)	Income from the Market (\$)	Degradation Cost (\$)	EVA Profit (\$)
60	1656	336	1320
30	1647	316	1331
15	1656	333	1323
5	1649	333	1316

TABLE V
IMPACT OF DIFFERENT REGULATION SIGNAL SCENARIO
GRANULARITIES

Scenario Granularity	EVA Profit (\$)	Solution Time - Bidding Problem (s)	Solution Time - Power Allocation Problem (s)
0.2	1330	1.63	0.15
0.1	1331	3.52	0.17
0.05	1336	8.80	0.14
0.025	1335	28.10	0.14
0.01	1334	81.21	0.16

Assumption 1. The bidding problem (16) and the power allocation problem (17) are strictly feasible.

Assumption 2. The actual regulation signals can be expressed by the regulation signal scenarios, i.e., $\forall \delta_{\hat{s}} \in [-1, 1]$, and we have $\hat{s} \in \hat{S}$ and $\pi_{t,\hat{s}} > 0, \forall t$.

Assumption 3. The occurrence frequency of the regulation signals is consistent with the forecasted probability.

Although our assumptions are not all true, the proposed framework works well!



Conclusion

1. An **EVA's bidding in the regulation market is coupled with its power allocation strategy** for regulation deployment. Considering this coupling improves the EVA profit.
2. The simulation results verify that co-optimizing bidding and power allocation enables both utilizing the **heterogeneous characteristics** of the EVs and **balancing the short/long-term costs**, thus improving the profit of the EVA while reducing the degradation costs.
3. The profit distribution between the EVA and EVs is also coupled with bidding and power allocation. In addition, the forecast of regulation signals could be improved. These issues are for future research.

Thanks for Listening!

Q & A

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